NetGuardAI

FYP-1 | Module 1 Planning Document | Prepared for Supervisor Meeting

FAST-NUCES | October 2026
Bachelor of Computer Science

Mid-Evaluation Preparation | Proposed Base Paper: Debelie et al. (2026) | Dataset: GeNIS (2025)

PROPOSED BASE PAPER	DATASET	CORE METHOD	PHASE	GOAL
Debelie et al. (2026)	GeNIS (2025)	CTGAN (To Be Evaluate	Module 1 - FYP-1	Mid-Evaluation

01. PROJECT OVERVIEW

NetGuardAI is a network intrusion detection system (IDS) that investigates whether generative AI techniques can improve the detection of underrepresented attack classes in imbalanced datasets.
Note: The objective is improving detection of known but rare attacks, not completely unseen attack types.

Our proposed base paper is Debelie et al. (2026), which studied GAN-based augmentation for intrusion detection. We will verify the exact methods their paper compares through a full literature review before finalising our research gap. Our work applies a similar methodology on the GeNIS (2025) dataset and investigates multiple augmentation strategies in a controlled, fair comparison. The goal is to determine which method, if any, improves detection of underrepresented attack classes on real held-out test data.

02. RESEARCH OBJECTIVE

Compare different data augmentation methods and determine which method, if any, improves minority-class intrusion detection on the GeNIS (2025) dataset.

This includes:

- Evaluating augmentation quality using both statistical measures (MMD, feature distributions) and downstream detection performance on real, held-out test data.
- Investigating whether synthetic training data improves detection of underrepresented attack classes without degrading performance on majority classes.
- Identifying which combination of augmentation method and classifier yields the best results under the same evaluation procedure.

Note: We do not assume any method will outperform others. Results are unknown until experiments are run. CTGAN, SMOTE, VAE, and GAN variants are all treated as candidates to be investigated.

03. SYSTEM MODULE OVERVIEW

NetGuardAI is divided into 5 modules. Modules 1-2 are targeted for FYP-1. Modules 3-5 are completed in FYP-2.

Module	Name	What It Does	Phase
M1	**Data Ingestion & Preprocessing**	**Load GeNIS, run EDA, clean data, apply correct preprocessing pipelin**	**FYP-1 (Now)**
M2	**Augmentation Comparison Study**	**Systematically compare augmentation methods using the same classi**	**FYP-1 (Start)**
M3	Synthetic Data Quality Evaluation	Evaluate synthetic data using MMD, feature distributions, correlations, and T	FYP-2
M4	Extended Classifier Evaluation	Add XGBoost and other classifiers after RF baseline if time and results perm	FYP-2
M5	SOC Dashboard Deployment	Build a real-time IDS dashboard as a deployable computing artifact	FYP-2

04. MODULE 1 - DETAILED TASK BREAKDOWN

Module 1 establishes the data foundation for the entire pipeline. A critical requirement is that the dataset is split BEFORE any preprocessing transformers are fitted. Encoders, scalers and

transformations must be fitted on training data only, then applied to validation and test sets.
Augmentation is applied to the training set only. Validation and test sets remain real and untouched.

#	Task Name	What We Will Do	Output
T1.1	**Dataset Acquisition**	Download GeNIS (2025) from Zenodo (CC BY 4.0 open access). Use the CSV version with flow-based records. Document the source, version, and number of flows obtained.	*GeNIS CSV files ready*
T1.2	**Data Loading & Inspection**	Load in Python with Pandas. Record total rows, columns, data types, and flow of each network scenario. Inspect for obvious quality issues.	*Dataset structure report*
T1.3	**Exploratory Data Analysis (ED**	Generate: (1) per-class sample count chart, (2) missing value heatmap, (3) feature correlation matrix, (4) summary statistics. Check for temporal or scenario groupings that could cause data leakage.	*4 to 5 charts for report*
T1.4	**Stratified Train/Val/Test Split**	Split FIRST before any preprocessing. Use 70/15/15 split. Check whether flows from the same capture or scenario appear across splits. If so, use a grouped or temporal split to prevent leakage.	*3 separate split files*
T1.5	**Feature Engineering - Training**	On training split only: remove irrelevant columns, handle missing values, encode categoricals. Fit all encoders on training data. Apply the fitted encoders to validation and test sets without re-fitting.	*Fitted encoders saved*
T1.6	**Normalization - Training Set O**	Fit scaler (e.g. Min-Max) on training set only. Transform all 3 splits using the training-fit scaler. Do NOT refit on validation or test data. This prevents data leakage.	*Scaler saved, 3 splits norma*
T1.7	**Class Imbalance Report**	Count samples per attack class in training set. Identify underrepresented classes that are candidates for augmentation. This documents the problem that motivates the study.	*Imbalance report + chart*
T1.8	**Baseline Model - No Augment**	Train Random Forest on the raw (non-augmented) training split. Evaluate on the real held-out test set. Record F1-score and Recall per class and report the actual results without prediction.	*Actual baseline results table*

05. AUGMENTATION COMPARISON STUDY PLAN (MODULE 2)

We will systematically compare augmentation methods under a controlled, fair experimental setup.
The same classifier (Random Forest) and the same evaluation procedure will be used for all methods.
Only the augmentation method changes between experiments. All augmentation is applied to the
training set only. The validation and test sets always contain real data.

Before listing comparison methods: confirm through a full reading of Debelie et al. (2026) which augmentation methods they actually compare. Do not claim the paper excludes SMOTE or VAE until this is verified. The research gap must be confirmed through the literature review.

Method	Type	Brief Description	Status
No Augmentation	**Baseline**	**Train on raw imbalanced training data. Evaluate on real test set. This is the co**	**Control**
SMOTE	Traditional	Oversample minority classes by interpolating between real minority samples. Applie	To Investigate
VAE	Deep Generative	Train a Variational Autoencoder per minority class on training data. Generate synthe	To Investigate
Vanilla GAN	Adversarial	Train a basic GAN per minority class. Generator learns to produce realistic samples	To Investigate
WGAN-GP	Adversarial	Wasserstein GAN with gradient penalty. More stable training than Vanilla GAN.	To Investigate
CTGAN	Adversarial	Conditional Tabular GAN, designed specifically for mixed-type tabular data. Applied	To Investigate

Outcomes are to be investigated, not assumed. No method is predicted to win or lose in advance.
Results will be reported as actual measured values from experiments.

06. SYNTHETIC DATA QUALITY EVALUATION PLAN

Synthetic data quality will be evaluated on two levels: statistical similarity and detection utility.

Evaluation Dimension	Method	What It Checks
Statistical Quality	MMD (Maximum Mean Discrepa	Measures distance between real and synthetic data distributions
Statistical Quality	Feature Distribution Comparisor	Compare histograms and summary stats of each feature: real vs synthetic
Statistical Quality	Correlation Structure	Check whether synthetic data preserves feature correlations found in real data
Detection Utility	Train on Augmented, Test on Re	Train classifier on augmented data, evaluate on real held-out test set
Detection Utility	Per-Class F1 and Recall	Check if augmentation actually improves minority-class detection on real data

07. CORRECT DATA PREPROCESSING WORKFLOW

The preprocessing pipeline must follow this strict order to prevent data leakage:

Step 1: Load raw GeNIS CSV data.
Step 2: Perform EDA on the raw, unsplit data (observation only, no transformation).
Step 3: Split data into Train / Validation / Test (70 / 15 / 15).
- Check if flows from the same capture or scenario span splits.
- Use a grouped or temporal split if leakage risk is identified.
Step 4: Fit all encoders and scalers on the TRAINING SET ONLY.
Step 5: Apply the fitted transformations to validation and test sets (no re-fitting).
Step 6: Apply augmentation (SMOTE, VAE, GAN, CTGAN, etc.) to the TRAINING SET ONLY.
Step 7: Validation and test sets remain real and untouched throughout all experiments.

This ensures the model is never exposed to any information from the validation or test sets during training or preprocessing, which is a fundamental requirement of honest evaluation.

08. CLASSIFIER PLAN

Classifier approach for this study:

Phase 1 (Module 2 - Augmentation Comparison):
- Use Random Forest as the single common classifier for ALL augmentation comparisons.
- This isolates augmentation as the only variable, making the comparison scientifically fair.

Phase 2 (FYP-2 - Extended Evaluation, time permitting):
- After identifying the best augmentation method, test additional classifiers (e.g. XGBoost) to see whether the improvement holds across different model families.
- Additional classifiers will only be added if time permits and there is a clear reason.

Note: We do not plan to train nine classifiers simultaneously. Classifier expansion is conditional on completing the augmentation comparison first.

09. 6-WEEK EXECUTION SCHEDULE

WEEK 1	WEEK 2	WEEK 3	WEEK 4	WEEK 5	WEEK 6
Download GeNIS dataset	Full EDA on raw data (T1	Perform data split (T1.4)	Class imbalance report (T	Read full Debelie et al. (2	Supervisor review
Setup environment	Check for leakage risk	Fit encoders on train only	Run baseline RF model (T	Confirm research gap	Fix all feedback
Inspect raw data (T1.1-T1	Generate all charts	Normalize correctly (T1.5	Report actual results	Begin SMOTE baseline	Prepare mid-eval slides

10. WHAT WE WILL SHOW AT MID-EVALUATION

[1] System Architecture Diagram

Full 5-module pipeline with FYP-1 / FYP-2 boundary and correct preprocessing flow labelled

[2] EDA Charts & Class Imbalance Report

Visual documentation of GeNIS dataset structure, class distribution, and identified leakage risks

[3] Correctly Preprocessed Dataset

Evidence that split was performed first and that transformers were fitted on training data only

[4] Actual Baseline Results

Real measured F1-score and Recall per class from RF trained on non-augmented training data

[5] Verified Research Gap

Evidence from full reading of Debelie et al. (2026) confirming which methods they compared

[6] Augmentation Comparison Design

Controlled experimental setup: same classifier (RF), same evaluation procedure, training-only augmentation